

Cross sections for $2 \rightarrow 2$ scattering

[1.] To calculate the cross section for $\psi\psi \rightarrow \chi\chi$ scattering, we evaluate the probability amplitude

$$T_{\chi\chi \leftarrow \psi\psi} = \langle f | T | i \rangle,$$

where

$$|i\rangle = \hat{a}_\psi^\dagger(\vec{p}_1) \hat{a}_\psi^\dagger(\vec{p}_2) |0\rangle \quad \text{and} \quad \langle f| = \langle 0 | \hat{a}_\chi(\vec{k}_1) \hat{a}_\chi(\vec{k}_2).$$

The interaction term in the Lagrangian is given by $\mathcal{L}_{\text{int}} = -\frac{1}{2}g'\phi\psi^2 - \frac{1}{2}g\phi\chi^2$. From this, we see that the first-order perturbation does not contribute to the probability amplitude since neither of the terms in $\mathcal{L}_{\text{int}}$ contains both ψ and χ which could be contracted with the creation / annihilation operators of the incoming / outgoing particles.

The lowest order in the couplings which allows the desired interaction to occur is the second order given by

$$T^{(2)} = \frac{i^2}{2!} \int d^4x d^4y \langle f | T [\mathcal{L}_{\text{int}}(x) \mathcal{L}_{\text{int}}(y)] | i \rangle.$$

Specifically, only the following term in the second perturbative order will contribute to the amplitude

$$T^{(2)} = \frac{i^2}{2!} \left(-\frac{g'}{2} \right) \left(-\frac{g}{2} \right) \int d^4x d^4y \langle f | T \left[\overline{\phi(x)\psi^2(x)\phi(y)\chi^2(y)} + \overline{\phi(y)\psi^2(y)\phi(x)\chi^2(x)} \right] | i \rangle,$$

as only this term allows for both the ϕ field to contract with itself and the ψ and χ fields to contract with the ingoing / outgoing particles.

Using the Feynman diagram rules in momentum space, we construct the following diagram as the sole second-order contribution to the connected T-matrix for the scattering process under consideration:

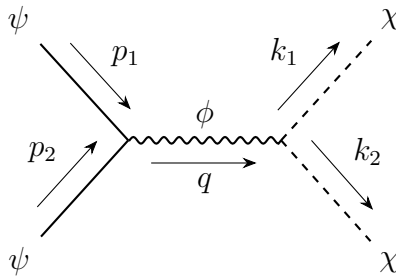


Figure 1: Feynman graph of $\psi\psi \rightarrow \chi\chi$ scattering

By reading this graph we obtain the following relation for the smooth connected T-matrix element:

$$i\mathcal{T}_C^{(2)} = i^2 g' g \frac{i}{(p_1 + p_2)^2 + i0^+}$$

The symmetry factor of the graph is $\mathcal{S} = 1$. In its computation we have omitted the 2 factor corresponding to the exchange $x \leftrightarrow y$ since the contributing term in the four-point function is already symmetrical to the exchange. Here, we directly substitute the Mandelstam invariant $s = (p_1 + p_2)^2$, reflecting that the studied process occurs only through the s -channel.

Now, we use the formula obtained for the $2 \rightarrow 2$ particle scattering

$$\frac{d\sigma}{d\Omega_{\text{CMS}}} = \frac{1}{64\pi^2} \frac{1}{s} \frac{|\vec{P}_{\text{CMS}}^f|}{|\vec{P}_{\text{CMS}}^i|} |\mathcal{T}_C|^2$$

In the CMS frame, the momenta can be expressed using the Mandelstam invariant s

$$\begin{aligned} p_1 &= \left(\frac{\sqrt{s}}{2}, \frac{\sqrt{s}}{2} \vec{e}_i\right), & k_1 &= \left(\frac{\sqrt{s}}{2}, \sqrt{\frac{s}{4} - m^2} \vec{e}_f\right), \\ p_2 &= \left(\frac{\sqrt{s}}{2}, -\frac{\sqrt{s}}{2} \vec{e}_i\right), & k_2 &= \left(\frac{\sqrt{s}}{2}, -\sqrt{\frac{s}{4} - m^2} \vec{e}_f\right). \end{aligned}$$

Substituting this into the formula for the differential cross section yields

$$\frac{d\sigma}{d\Omega_{\text{CMS}}} = \frac{1}{64\pi^2} \frac{1}{s} \sqrt{1 - \frac{4m^2}{s}} |\mathcal{T}_C|^2$$

Finally, we substitute the relation for the smooth connected T-matrix element for the lowest order in the couplings

$$\frac{d\sigma}{d\Omega_{\text{CMS}}} \approx \frac{1}{64\pi^2} \frac{1}{s} \sqrt{1 - \frac{4m^2}{s}} \frac{(g'g)^2}{s^2}$$

Since the Mandelstam invariant s does not depend on the ϑ angle, the integration over $d\Omega_{\text{CMS}}$ is trivial and yields only the factor 4π . However, we must also add the factor $\frac{1}{2}$ which compensates the fact that the final states $|\vec{P}_{\text{CMS}}^f, -\vec{P}_{\text{CMS}}^f\rangle$ and $|- \vec{P}_{\text{CMS}}^f, \vec{P}_{\text{CMS}}^f\rangle$ are indistinguishable while the integration should be over distinct final states only. In conclusion, we derive the following expression for the total cross section at the lowest order in couplings:

$$\sigma \approx \frac{1}{32\pi} \sqrt{1 - \frac{4m^2}{s}} \frac{(g'g)^2}{s^3}$$

[2.] In this problem, the procedure is analogous to the previous calculation. We again want to compute the probability amplitude

$$T_{\psi\psi \leftarrow \psi\psi} = \langle f | T | i \rangle,$$

where

$$|i\rangle = \hat{a}_\psi^\dagger(\vec{p}_1) \hat{a}_\psi^\dagger(\vec{p}_2) |0\rangle \quad \text{and} \quad \langle f| = \langle 0 | \hat{a}_\psi(\vec{k}_1) \hat{a}_\psi(\vec{k}_2)$$

with the same interaction Lagrangian $\mathcal{L}_{\text{int}} = -\frac{1}{2}g'\phi\psi^2 - \frac{1}{2}g\phi\chi^2$. In this case, we would get a non-zero amplitude even in the zeroth order, however, the cross section is only given by the connected part of the S-matrix elements, or the connected part of the T-matrix to be more precise, and therefore this non-zero amplitude does not interest us. The first perturbative order does not contribute to the probability amplitude for the same reason as in the previous case.

Again, the lowest order in the coupling which allows the scattering to occur is the second order. In this case, the contributing term is the following

$$T^{(2)} = \frac{i^2}{2!} \left(-\frac{g'}{2} \right)^2 \int dx^4 dy^4 \langle f | T \left[\overline{\phi(x)\psi^2(x)} \phi(y)\psi^2(y) \right] | i \rangle$$

In this case however, three Feynman graphs with unique topology can be drawn:

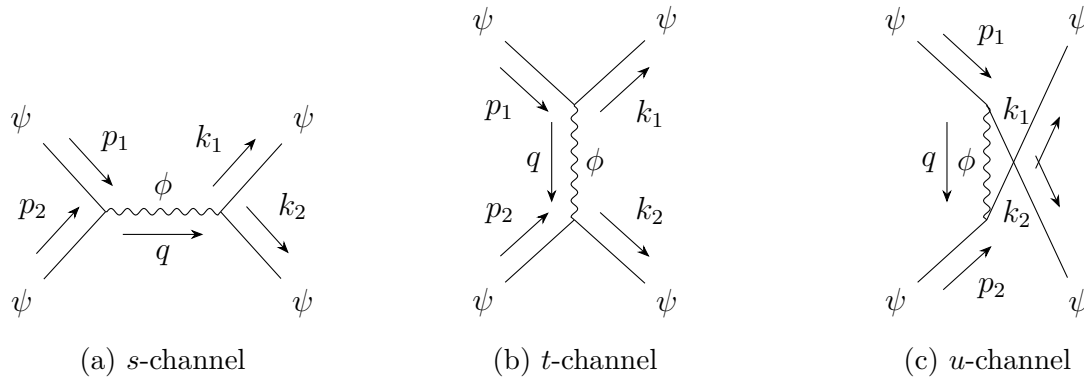


Figure 2: Feynman graph of $\psi\psi \rightarrow \psi\psi$ scattering

By reading these graphs, which again have the symmetry factor equal to one, we obtain the smooth connected T-matrix element which composes of the sum of three factors corresponding to *s*-, *t*- and *u*-channel respectively:

$$\begin{aligned} i\mathcal{T}_C^{(2)} &= i\mathcal{M}_s + i\mathcal{M}_t + i\mathcal{M}_u \\ &= i^2(g')^2 \left[\frac{i}{(p_1 + p_2)^2 + i0^+} + \frac{i}{(p_1 - k_1)^2 + i0^+} + \frac{i}{(p_1 - k_2)^2 + i0^+} \right] \end{aligned}$$

Here, we can again directly substitute the Mandelstam invariants $s = (p_1 + p_2)^2$, $t = (p_1 - k_1)^2$ and $u = (p_1 - k_2)^2$ respectively.

The formula for the differential cross section in the case of $2 \rightarrow 2$ particle scattering simplifies since the $|\vec{P}_{\text{CMS}}^i|$ and $|\vec{P}_{\text{CMS}}^f|$ are equal in the case of identical particle scattering. Overall, we obtain the final relation for the differential cross section:

$$\frac{d\sigma}{d\Omega_{\text{CMS}}} \approx \frac{1}{64\pi^2} \frac{(g')^4}{s} \left[\frac{1}{s} + \frac{1}{t} + \frac{1}{u} \right]^2$$

Depending on the convention, an additional factor of $\frac{1}{2}$ may be included in the differential cross section to account for the indistinguishability of the particles.